PassFrame tuition

Roman Numerals & Negative Numbers

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. XXVI (XX + VI)

2. C — XXX + VIII = 30 + 8 = 38

3. XLVII (XL + VII)

4. 52 (L + II)

5. B — XC = C − X = 90

6. -7°C (4 − 11 = −7)

7. LXIII (LX + III)

8. -14°C, -9°C, -2°C, 3°C

9. B, D and E (LXII=62, LXXIV=74 and LVIII=58 are greater than 50; XLIX=49 and XLV=45 are not)

10. 84 (LXXX + IV = 80 + 4)

11. 5°C (−8 + 13 = 5)

12. D — -15°C is the lowest value

13. LXXII (LXX + II)

14. 99 (XC + IX = 90 + 9)

15. -15°C (−3 − 12 = −15)

Standard

1. XCI (XC + I = 90 + 1)

2. 89 (LXXX + IX = 80 + 9)

3. B — XC + VI = 90 + 6 = 96

4. -8°C (7 − 15 = −8)

5. LXVIII (LX + VIII = 60 + 8)

6. -20, -12, -1, 4

7. C — XC + VII = 90 + 7 = 97

8. 6°C (−14 + 20 = 6)

9. LXXVI (LXX + VI = 70 + 6)

10. 17°C (8 − (−9) = 17)

11. B, C and E (LXXIII=73, LXXXVIII=88 and LXVII=67 are all between 60 and 90; LIV=54 is below 60 and XCII=92 is above 90)

12. XLIII (XL + III = 40 + 3)

13. 87 (LXXX + VII = 80 + 7)

14. C — 5°C is the highest value

15. -6°C (−9 − 8 = −17; −17 + 11 = −6)

Standard / Hard

1. XCVII (XC + VII = 90 + 7)

2. C — XC + VIII = 90 + 8 = 98

3. 17°C (9 − (−8) = 17)

4. LXXIX (LXX + IX = 70 + 9)

5. -17 m (−24 + 16 = −8; −8 − 9 = −17)

6. A — 60 + 30 = 90 = XC

7. LXXXIV (LXXX + IV = 80 + 4)

8. 6°C, 1°C, -3°C, -19°C

9. A, C and E (XCIX=99, −17°C is colder than −12°C, and LXVI=66 are true; LXXIV is actually 74 (not 84) and −4°C is colder than 2°C (not warmer))

10. 93 (XC + III = 90 + 3)

11. Floor -3 (3 − 8 = −5; −5 + 2 = −3)

12. B — −12 + 7 = −5; −5 + 9 = 4

13. LXXXIX (LXXX + IX = 80 + 9)

14. 66 (XCVI = 96; 96 − 30 = 66)

15. 7°C (−18 + 25 = 7)

Homework — Answers

Easy / Standard

1. XXII (XX + II)

6. -7°C (5 − 12 = −7)

11. 3°C (−7 + 10 = 3)

2. C — XXX + IV = 30 + 4 = 34

3. XXXIX (XXX + IX)

4. 66 (LX + VI = 60 + 6)

5. A — XL + IV = 40 + 4 = 44

7. LV (L + V)

8. -10°C, -6°C, -1°C, 2°C

9. B, C and E (XLVI=46, LIII=53 and XLI=41 are greater than 40; XXXVII=37 and XXXIV=34 are not)

10. 73 (LXX + III = 70 + 3)

12. C — -13°C is the lowest value

13. LXI (LX + I)

14. 49 (XL + IX = 40 + 9)

15. -11°C (−2 − 9 = −11)

Standard

1. LXXXII (LXXX + II = 80 + 2)

2. 68 (LX + VIII = 60 + 8)

3. A — XC + IV = 90 + 4 = 94

4. -9°C (8 − 17 = −9)

5. XXXVII (XXX + VII)

6. -22, -15, -3, 6

7. C — LXXX + I = 80 + 1 = 81

8. 6°C (−19 + 25 = 6)

9. LXXXVII (LXXX + VII)

10. 17°C (6 − (−11) = 17)

11. B, C and E (LXVII=67, LXXVIII=78 and LIII=53 are all between 50 and 80; XLVIII=48 is below 50 and XC=90 is above 80)

12. XLVI (XL + VI)

13. 92 (XC + II = 90 + 2)

14. C — 4°C is the highest value

15. -3°C (−6 − 9 = −15; −15 + 12 = −3)

Standard / Advanced

1. 83 (LXVIII = 68; 68 + 15 = 83)

2. B — 3 − 8 = −5°C

3. 74 = LXXIV; 20 less is 54 = LIV

4. 72 (XXXVI = 36; 36 × 2 = 72)

5. C — −9 + 14 = 5°C

6. 62 = LXII; 62 + 25 = 87 = LXXXVII

7. 24°C (13 − (−11) = 24)

8. A, C, D and E (LXXXI=81; −13°C is 5°C colder than −8°C (−8−5=−13); 4−(−11)=15 so 4°C is 15°C warmer; XLVIII=48 — all correct; XCVII is actually 97, not 87)

9. LXXXIV = 84; 84 − 25 = 59 = LIX

10. A — −22 + 9 = −13; −13 − 15 = −28

11. 93 = XCIII; 93 − 47 = 46 = XLVI

12. 10°C to -8°C is a drop of 18°C; half of 18 is 9; -8 + 9 = 1°C

13. C — LXXXVIII = 88; 88 × 2 = 176

14. £33 (−56 + 89 = 33)

15. 78 = LXXVIII, 41 = XLI; difference = 78 − 41 = 37